

# Affordance Segmentation Network for Multi-Instance Environments

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## INTRODUCTION

Most existing affordance studies assume a single object per image, while real-world environments typically contain multiple instances of the same object with potentially different affordances. This mismatch limits the applicability of prior methods in realistic scenarios such as robotic manipulation and scene understanding.

To address this limitation, we propose **Multi-AffordanceNet**, a framework that predicts affordances at the instance level, explicitly separating affordance predictions for each object instance in an image. We further construct a multi-instance affordance dataset using a copy-and-paste strategy and design an instance-aware training

## PROBLEM DEFINITION

- Affordance Prediction Objective
  - Affordance prediction aims to identify actionable regions in an image that indicate how an agent can interact with objects.
- Ambiguity in Multi-Instance Scenes
  - In real-world images, multiple instances of the same object category often appear simultaneously.
  - Without instance-level separation, affordance predictions become ambiguous, making it unclear which object instance affords which action.



- Multi-Instance Affordance Prediction Goal
  - detect all object instances present in the scene, and
  - predict an affordance map for each instance independently.

## METHOD

- Dataset Creation
  - copy-paste



- mosaic



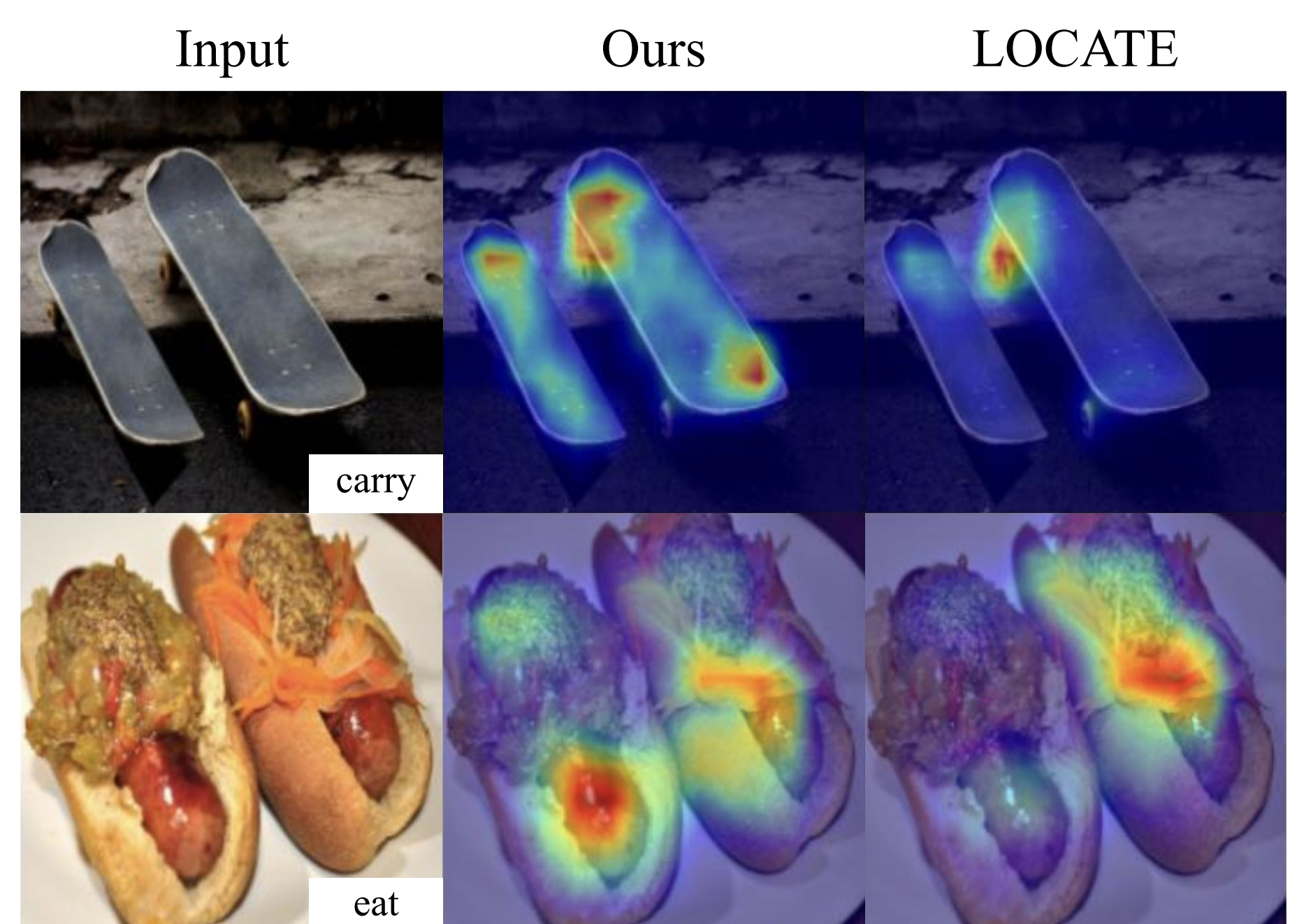
- Consistency Regularization

$$\mathcal{L}_{total} = \mathcal{L}_{LOCATE}(x_{single}) + \mathcal{L}_{LOCATE}(x_{copy-paste}) + \lambda \cdot \mathcal{L}_{consistency}$$

$$\mathcal{L}_{LOCATE} = \alpha \cdot \mathcal{L}_{KLD} + \beta \cdot \mathcal{L}_{SIM} + \gamma \cdot \mathcal{L}_{NSS}$$

$$\mathcal{L}_{consistency} = \|\mathcal{T}(f(x_{single})) - f(\mathcal{T}(x_{single}))\|^2$$

## RESULTS



## CONCLUSION

Multi-AffordanceNet effectively addresses multi-instance affordance prediction in complex scenes. However, challenges remain in accurately predicting affordances under object occlusion. Future work will focus on occlusion-aware representations and extend the model toward spatial affordance reasoning that captures interactions and relationships among objects.